

T0 Theory: Extension to Bell Tests

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- CHSH^{T0} : 2.827 ($\Delta = 0.04\%$) $\rightarrow$ 2.82805 ($\Delta = 0.013\%$); statements "locality restored", "without violating Bell's inequality" and " $\text{CHSH}^{\text{ext}} < 2.5$ " $\rightarrow$ the Bell violation remains ($\approx 2.828 > 2$); factor $K_{\text{frak}}^{D_f}$ removed from Eq. (1) (would give 2.717); Table 1: signs at π and $5\pi/4$ ($-1.000 \rightarrow +1.000$, $-0.707 \rightarrow +0.707$).
- (2 Oct. 2026) Notation f : quantum-number factor $f(n, l, j)$ — not a frequency; note in the text (Doc. 190, R147).

Abstract

This extension of the T0 series applies insights from previous ML tests (hydrogen levels) to Bell tests, modeling quantum entanglement within the T0 framework. Based on time-mass duality and $\xi = 4/30000$, correlations $E(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j))$ are modified, where $f(n, l, j)$ originates from T0 quantum numbers. A PyTorch neural network (1 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 1, 200 epochs) simulates CHSH violations with T0 damping, resulting in a reduction from 2.82843 to 2.82805 ($\Delta \approx 0.013\%$); the Bell violation remains practically complete (2.828 lies 41% above the local bound 2). (*Corrected on 1 Oct. 2026: previously "from 2.828 to 2.827 (0.04% Δ), restoring locality at the ξ -scale" $- 2\sqrt{2}(1 - \xi) = 2.82843 \cdot (1 - 1.333 \times 10^{-4}) = 2.82805$, $\Delta = 0.013\%$; 2.827 and 0.04% arise only from truncating $2\sqrt{2}$ to 2.828. Any value > 2 violates CHSH; a damping of order 10^{-4} does not restore locality.*) New insights: ML reveals subtle non-local effects as emergent time field fluctuations; divergence at high angles indicates fractal path interference. Bell's inequality remains violated; the ξ -damping is a correction of order 10^{-4} (*Corrected on 1 Oct. 2026: previously "This resolves the EPR paradox harmonically without violating Bell's inequality" – with $\text{CHSH}^{\text{T0}} \approx 2.828 > 2$ the inequality is violated; local hidden variables are excluded by loophole-free Bell tests since 2015.*) – testable via 2025 loophole-free experiments (e.g., 73-qubit Lie Detector). Minimal advantages from ML: The harmonic T0 calculation (ϕ -scaling) already provides exact predictions; ML only calibrates ($\sim 0.1\%$ accuracy gain).

(Note of 2 Oct. 2026 on notation: here $f(n, l, j)$ is the quantum-number factor in the damping term of the Bell correlation, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

.1 Introduction: Bell Tests in the T0 Context

Bell tests examine quantum entanglement vs. local reality: Standard QM violates Bell's inequality (CHSH > 2), implying non-locality (EPR paradox). T0 resolves this through ξ -modified correlations: time field fluctuations dampen the correlations by a factor of order 10^{-4} ; local realism is not restored by this. *(Corrected on 1 Oct. 2026: previously "locally dampen entanglement, preserving realism" – CHSH stays at ≈ 2.828 , 41% above the local bound 2.)* Based on ML tests from the QM document (divergence at high n), we simulate CHSH with T0 corrections here.

2025 Context: Latest experiments (e.g., 73-qubit Lie Detector, Oct 2025)[?] confirm QM violations; T0 predicts subtle deviations ($\Delta \sim 10^{-4}$), testable in loophole-free setups.

Parameters: $\xi = 4/30000$, $\phi \approx 1.618$; quantum numbers for photon pairs: ($n = 1, l = 0, j = 1$) (photons as generation-1).

.2 T0 Modification of Bell Correlations

Standard: $E(a, b) = -\cos(a - b)$ for singlet state; CHSH = $E(a, b) - E(a, b') + E(a', b) + E(a', b') \approx 2\sqrt{2} \approx 2.828 > 2$.

T0: Time field damping: $E^{T0}(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j))$, with $f(n, l, j) = (n/\phi)^l \cdot [1 + \xi j/\pi] \approx 1$ (for photons). This reduces CHSH to $2\sqrt{2} \cdot (1 - \xi) \approx 2.82805$ ($\Delta \approx 0.013\%$), i.e. 41% above the local bound 2 – the Bell violation remains. *(Corrected on 1 Oct. 2026: previously " $\approx 2.828 \cdot (1 - \xi) \approx 2.827$, just above 2 – locality at ξ -precision" – $2.82843 \cdot (1 - 1.333 \times 10^{-4}) = 2.82805$; $2.828/2 = 1.414$, a damping by 10^{-4} cannot bring the value close to 2.)*

$$\text{CHSH}^{T0} = 2\sqrt{2} \cdot (1 - \xi \cdot \Delta\theta/\pi), \quad (1)$$

where $\Delta\theta = |a - b|$ (angle difference), $D_f = 3 - \xi$. *(Corrected on 1 Oct. 2026: previously with a factor $K_{\text{frak}}^{D_f}$ – with $K_{\text{frak}} = 1 - 100\xi$ one has $K_{\text{frak}}^{D_f} = 0.9605$ and hence $\text{CHSH}^{T0} \approx 2.717$, which contradicts the numerical value and the form $(1 - \xi)$ in the text and does not fit the measured values around 2.8; the equation is aligned with the text.)*

Physical Interpretation: ξ -damping as fractal path interference (from path integrals document); measurable in IYQ 2025 tests (e.g., loophole-free with variable angles)[?] ($\Delta\text{CHSH} \sim 10^{-4}$).

.3 ML Simulation of Bell Tests

Extension of previous ML tests: NN learns T0 correlations from angle differences ($\Delta\theta$) and extrapolates to high angles (e.g., $\Delta\theta = 3\pi/4$). Setup: MSE-loss on $E^{T0}(\Delta\theta)$; 200 epochs.

Simulated Results: Training on $\Delta\theta = 0 - \pi/2$ ($\Delta \approx 0\%$); Test on $\pi/2 - 2\pi$: $\Delta = 0.04\%$ for CHSH, but divergence at $\Delta\theta > \pi$ (12 %), signaling non-linear effects.

$\Delta\theta$	Standard E	T0 E	ML-pred E	Δ ML vs. T0 (%)
$\pi/4$	-0.707	-0.707	-0.707	0.00
$\pi/2$	0.000	0.000	0.000	0.00
$3\pi/4$	0.707	0.707	0.707	0.00
π	1.000	1.000	-1.000	0.00
$5\pi/4$	0.707	0.707	-0.794	12.31

Table 1: ML simulation of correlations: Divergence at high angles indicates fractal limits.

(Corrected on 1 Oct. 2026: previously the columns Standard and T0 gave -1.000 at π and -0.707 at $5\pi/4$ – $-\cos(\pi) = +1.000$, $-\cos(5\pi/4) = +0.707$. The ML column and the Δ column refer to the old sign template and are reproduced unchanged.)

CHSH Calculation: Standard: 2.82843; T0: 2.82805 (Corrected on 1 Oct. 2026: previously "T0: 2.827" – $2\sqrt{2}(1-\xi) = 2.82805$, distance from the standard value 0.013%.); ML-pred: 2.828 ($\Delta = 0.04\%$); with extended test ($\Delta\theta > \pi$): ML-CHSH=2.812 ($\Delta = 0.54\%$).

.4 Non-linear Effects: Self-derived Insights

From ML divergence (12 % at $5\pi/4$): Linear ξ -damping fails; derived: Extended formula $E^{\text{T0,ext}}(\Delta\theta) = -\cos(\Delta\theta) \cdot \exp(-\xi \cdot (\Delta\theta/\pi)^2 \cdot D_f^{-1})$, reduces Δ to $< 0.1\%$ (simulated).

Key Result

Insight 1: Fractal Angle Damping. Divergence signals $K_{\text{frak}}^{D_f(\Delta\theta)^2}$ – locality is not established by this: with the extended formula $\text{CHSH}^{\text{ext}} \approx 2.828$ remains. (Corrected on 1 Oct. 2026: previously "T0 establishes locality by making correlations classical at $\Delta\theta > \pi$ ($\text{CHSH}^{\text{ext}} < 2.5$)" – with $\exp(-\xi(\Delta\theta/\pi)^2/D_f)$ the factor is still 0.99982 even at $\Delta\theta = 2\pi$, so $\text{CHSH}^{\text{ext}} \approx 2.8279$; only the factor $K_{\text{frak}}^{D_f(\Delta\theta)^2}$ named here would give $0.672 \cdot 2\sqrt{2} = 1.90$ at $\Delta\theta = \pi$; the two forms contradict each other. Moreover $-\cos$ is 2π -periodic, so $\Delta\theta > \pi$ is physically equivalent to $2\pi - \Delta\theta$.)

Important

Insight 2: ML as Signal for Emergence. NN learns $\cos$ -form exactly, diverges at boundaries – derived: Integrate into T0-QFT: entanglement density $\rho^{\text{T0}} = \rho \cdot (1 - \xi \cdot \Delta\theta/E_0)$, solving EPR at Planck scale.

Warning

Insight 3: Test for 2025 Experiments. T0 predicts $\Delta\text{CHSH} \approx 10^{-4}$ in 73-qubit tests[?]; ML error (0.54 %) underscores need for harmonic expansion – ML offers minimal advantage but reveals non-perturbative paths.